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Course : Programming in Java
Course code : CSA0974

1. Library Management System

Design a Java class to represent a library book.
Data Members:
• Book ID
• Book Title
• Author Name
• Number of Copies Available
Methods:
1. Read book details.
2. Issue a book (decrease available copies by 1).
3. Return a book (increase available copies by 1).
4. Display updated book details.
Assume: Initial copies = 20.

CODE :

```
import java.util.Scanner;
class LibraryBook {
    int bookId;
    String bookTitle;
    String authorName;
    int copies = 20;
    void readDetails() {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter Book ID: ");
        bookId = sc.nextInt();
        sc.nextLine();
        System.out.print("Enter Book Title: ");
        bookTitle = sc.nextLine();
        System.out.print("Enter Author Name: ");
        authorName = sc.nextLine();
    }
    void issueBook() {
        if (copies > 0) {
            copies--;
            System.out.println("Book Issued Successfully.");
        } else {
            System.out.println("No copies available.");
        }
    }
    void returnBook() {
        copies++;
        System.out.println("Book Returned Successfully.");
    }
```

```
    void displayDetails() {
        System.out.println("\nBook Details");
        System.out.println("Book ID: " + bookId);
        System.out.println("Book Title: " + bookTitle);
        System.out.println("Author Name: " + authorName);
        System.out.println("Available Copies: " + copies);
    }
}
public class LibraryManagement {
    public static void main(String[] args) {
        LibraryBook book = new LibraryBook();
        book.readDetails();
        book.issueBook();
        book.returnBook();
        book.displayDetails();
    }
}
```

INPUT :

101
Java Programming
James Gosling

OUTPUT :

Enter Book ID: 101
Enter Book Title: Java Programming
Enter Author Name: James Gosling
Book Issued Successfully.
Book Returned Successfully.
Book Details
Book ID: 101
Book Title: Java Programming
Author Name: James Gosling
Available Copies: 20

2. Employee Salary Management

Design a Java class to represent an employee.

Data Members:

• Employee ID
• Employee Name
• Department
• Basic Salary

Methods:

1. Read employee details.
2. Calculate HRA (20% of Basic Salary).
3. Calculate DA (10% of Basic Salary).
4. Display gross salary.

Assume: Basic Salary = ₹25,000.

CODE :

```
import java.util.Scanner;
class Employee {
    int empId;
    String empName;
    String department;
    double basicSalary = 25000;
    void readDetails() {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter Employee ID: ");
        empId = sc.nextInt();
        sc.nextLine();
        System.out.print("Enter Employee Name: ");
        empName = sc.nextLine();
        System.out.print("Enter Department: ");
        department = sc.nextLine();
    }
    void displayGrossSalary() {
        double hra = basicSalary * 0.20;
        double da = basicSalary * 0.10;
        double grossSalary = basicSalary + hra + da;
        System.out.println("\nEmployee Details");
        System.out.println("Employee ID: " + empId);
        System.out.println("Employee Name: " + empName);
        System.out.println("Department: " + department);
        System.out.println("Basic Salary: ₹" + basicSalary);
        System.out.println("HRA: ₹" + hra);
        System.out.println("DA: ₹" + da);
        System.out.println("Gross Salary: ₹" + grossSalary);
    }
}
public class EmployeeSalaryManagement {
    public static void main(String[] args) {
        Employee emp = new Employee();
        emp.readDetails();
        emp.displayGrossSalary();
    }
}
```

INPUT :

2005
Anitha
HR

OUTPUT :

Enter Employee ID: 2005
Enter Employee Name: Anitha
Enter Department: HR

Employee Details
Employee ID: 2005
Employee Name: Anitha
Department: HR
Basic Salary: ₹25000.0
HRA: ₹5000.0
DA: ₹2500.0
Gross Salary: ₹32500.0

3. Student Fee Management System
Design a Java class to represent a student.

Data Members:
• Student ID
• Student Name
• Course
• Fee Balance

Methods:
1. Read student details.
2. Pay fee amount.
3. Check remaining fee balance.
4. Display student details.

Assume: Total Course Fee = ₹50,000.

CODE:

```
import java.util.Scanner;
class Student {
    int studentId;
    String studentName;
    String course;
    double feeBalance;
    void readDetails() {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter Student ID: ");
        studentId = sc.nextInt();
        sc.nextLine();
        System.out.print("Enter Student Name: ");
        studentName = sc.nextLine();
        System.out.print("Enter Course: ");
        course = sc.nextLine();
        System.out.print("Enter Fee Balance: ");
        feeBalance = sc.nextDouble();
    }
    void payFee(double amount) {
```

```
        feeBalance = feeBalance - amount;
        System.out.println("Fee Paid: ₹" + amount);
    }
    void checkBalance() {
        System.out.println("Remaining Fee Balance: ₹" + feeBalance);
    }
    void displayDetails() {
        System.out.println("\nStudent Details");
        System.out.println("Student ID: " + studentId);
        System.out.println("Student Name: " + studentName);
        System.out.println("Course: " + course);
        System.out.println("Fee Balance: ₹" + feeBalance);
    }
}

public class StudentFeeManagement {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        Student s = new Student();
        s.readDetails();
        System.out.print("Enter Fee Amount to Pay: ");
        double amount = sc.nextDouble();
        s.payFee(amount);
        s.checkBalance();
        s.displayDetails();
    }
}
```

INPUT:

Enter Student ID: 205
Enter Student Name: Priya
Enter Course: B.Sc Computer Science
Enter Fee Balance: 40000
Enter Fee Amount to Pay: 12000

OUTPUT:

Fee Paid: ₹12000.0
Remaining Fee Balance: ₹28000.0
Student Details
Student ID: 205
Student Name: Priya
Course: B.Sc Computer Science
Fee Balance: ₹28000.0

4.Electricity Bill Management

Design a Java class to represent a consumer.

Data Members:

- Consumer Number
- Consumer Name
- Type of Connection (Domestic/Commercial)
- Units Consumed

Methods:

1. Read consumer details.
2. Calculate electricity bill.
3. Display bill amount.
4. Validate connection type.

Assume:

- Domestic: ₹5 per unit
- Commercial: ₹8 per unit

CODE :

```
import java.util.Scanner;
class Consumer {
    int consumerNumber;
    String consumerName;
    String connectionType;
    int unitsConsumed;
    double billAmount;
    void readDetails() {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter Consumer Number: ");
        consumerNumber = sc.nextInt();
        sc.nextLine();
        System.out.print("Enter Consumer Name: ");
        consumerName = sc.nextLine();
        System.out.print("Enter Connection Type (Domestic/Commercial): ");
        connectionType = sc.nextLine();
        System.out.print("Enter Units Consumed: ");
        unitsConsumed = sc.nextInt();
    }
    boolean validateConnectionType() {
        return connectionType.equalsIgnoreCase("Domestic") ||
               connectionType.equalsIgnoreCase("Commercial");
    }
    void calculateBill() {
        if (connectionType.equalsIgnoreCase("Domestic")) {
            billAmount = unitsConsumed * 5;
        } else if (connectionType.equalsIgnoreCase("Commercial")) {
            billAmount = unitsConsumed * 8;
        }
    }

    void displayBill() {
        System.out.println("\nConsumer Number: " + consumerNumber);
```

```
        System.out.println("Consumer Name: " + consumerName);
        System.out.println("Connection Type: " + connectionType);
        System.out.println("Units Consumed: " + unitsConsumed);
        System.out.println("Bill Amount: ₹" + billAmount);
    }
}
public class ElectricityBillManagement {
    public static void main(String[] args) {
        Consumer c = new Consumer();
        c.readDetails();
        if (c.validateConnectionType()) {
            c.calculateBill();
            c.displayBill();
        } else {
            System.out.println("Invalid Connection Type!");
        }
    }
}
```

INPUT:

Enter Consumer Number: 2005
Enter Consumer Name: Priya
Enter Connection Type (Domestic/Commercial): Commercial
Enter Units Consumed: 300

OUTPUT:

Consumer Number: 2005
Consumer Name: Priya
Connection Type: Commercial
Units Consumed: 300
Bill Amount: ₹2400.0

5. Mobile Recharge System

Design a Java class to represent a mobile subscriber.

Data Members:

- Mobile Number
- Subscriber Name
- Service Provider
- Account Balance

Methods:

1. Read subscriber details.
2. Recharge account balance.
3. Deduct balance for a call/data usage.
4. Display updated balance.

Assume: Initial balance = ₹500.

CODE :

```
import java.util.Scanner;
class Subscriber {
    String mobileNumber;
    String subscriberName;
    String serviceProvider;
    double accountBalance = 500;
    void readDetails() {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter Mobile Number: ");
        mobileNumber = sc.nextLine();
        System.out.print("Enter Subscriber Name: ");
        subscriberName = sc.nextLine();
        System.out.print("Enter Service Provider: ");
        serviceProvider = sc.nextLine();
    }
    void recharge(double amount) {
        accountBalance += amount;
        System.out.println("Recharge Successful: ₹" + amount);
    }
    void deductBalance(double amount) {
        accountBalance -= amount;
        System.out.println("Usage Deducted: ₹" + amount);
    }
    void displayBalance() {
        System.out.println("\nMobile Number: " + mobileNumber);
        System.out.println("Subscriber Name: " + subscriberName);
        System.out.println("Service Provider: " + serviceProvider);
        System.out.println("Updated Balance: ₹" + accountBalance);
    }
}
public class MobileRechargeSystem {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        Subscriber s = new Subscriber();
        s.readDetails();
        System.out.print("Enter Recharge Amount: ");
        double recharge = sc.nextDouble();
        s.recharge(recharge);
        System.out.print("Enter Usage Amount: ");
        double usage = sc.nextDouble();
        s.deductBalance(usage);

        s.displayBalance();
    }
}
```

INPUT:

Enter Mobile Number: 9876543210
Enter Subscriber Name: Arun
Enter Service Provider: Airtel
Enter Recharge Amount: 200
Enter Usage Amount: 50

OUTPUT :

Recharge Successful: ₹200.0
Usage Deducted: ₹50.0
Mobile Number: 9876543210
Subscriber Name: Arun
Service Provider: Airtel
Updated Balance: ₹650.0